Mohammed Salih

Backend-Focused Full-Stack Developer & QA Engineer
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Professional Summary

Backend-focused Full-Stack Developer and QA Engineer with 5 years of experience building production-ready web platforms and large-scale backend systems across full-time and independent work. Serve as CTO and Backend Lead at AfroDigital Innovation Labs, specializing in custom AI solutions. Built the backend for an integrated health service platform with a multi-vendor pharmacy marketplace, doctor consultations, and integrated digital payments, and for an AI-native school operating system that brings HR, finance, students, and parents into one platform with support for five languages. Own products end to end, mentoring and leading a team of 10 engineers delivering AI products for multiple clients. Founded BioMatch, an AI-powered organ donor matching platform recognized as a top startup at Lead Business 2025. Work mainly in TypeScript, Node.js (NestJS, Bun), and Python.

Experience

AfroDigital Innovation Labs PLC — afrodigital.dev 2022 – Present
CTO & Backend Lead *Full-time (Addis Ababa, Ethiopia)*

- Delivered backend and AI solutions for 8+ client organizations and private clients across education, healthcare, and marketing, including custom ERPs, AI chatbots, workflow automations, and rich frontend UIs
- Own the architecture for 5+ client and in-house products, choosing the tech stack, designing APIs and data models, and setting code review and deployment standards, while planning roadmaps closely with the CEO
- Authored and reviewed 70+ real-world software engineering tasks with reference solutions, used to fine-tune and evaluate custom AI models
- Cut average product shipping and delivery time from 6 weeks to 3 weeks by building a reusable NestJS backend foundation, now the company standard across client projects, and automating CI/CD (GitHub Actions, Docker)
- Lead backend developer for YeneSchool, an AI-Director-Driven School Operating System for Ethiopian schools: built 24+ modules across 7 user roles, including HR and payroll, fee management, attendance, grading and report cards, and parent communication, with an AI Director that automates routine admin work in five languages
- Lead developer for Tibeb, an integrated health platform in Addis Ababa: built a multi-vendor pharmacy marketplace where licensed pharmacies manage their own storefronts, inventory, and pricing, with prescription uploads, online doctor consultations, and automated vendor payouts
- Mentor and lead a team of 10 engineers developing 2 projects under a formal Agile lifecycle, using code reviews, architecture guidance, and pairing; onboard 5+ junior developers and give them day-to-day technical guidance

BITS College School of Systems and Technology 2025 – Present
Student Instructor & Backend Developer (Contract) *Part-time (Addis Ababa, Ethiopia)*

- Selected on the basis of academic performance (4.0 GPA) to teach junior software engineering students while still studying at the college
- Taught 40+ first-year students in the Python lab, guiding them from the concepts covered in lectures to writing working programs through hands-on coding practice
- Taught practical Data Structures & Algorithms to second-year students, guiding them to a deeper understanding of core concepts and stronger problem-solving skills
- Backend developer (contract) for Melue, a therapy management system digitizing ABA therapy for children with autism: built backend features for assessments, therapy plans, live session data collection, staff scheduling, and a parent portal (Ruby on Rails 8, PostgreSQL, Rodauth/JWT, RSpec)

Projects

Parsim | *Inference & Context Reduction for Long-Horizon Agents* – https://parsim.app/
Founder & Backend Engineer · Personal product, in development

Designing and building the backend for an inference layer for long-horizon AI agents, which waste tokens reprocessing history that is no longer relevant. Parsim compresses agent context while keeping task-relevant information, so agents run longer tasks with faster responses and lower token usage.

YeneSchool | *AI-Driven School Operating System* – https://yeneschool.me/

Schools run HR, finance, academics, and parent communication across separate tools. Lead development of an AI-driven operating system that brings every school stakeholder, including HR, finance, students, and parents, into one platform, with support for five languages and AI automation of routine school workflows.

Agentum | *AI-Native Customer Support* – https://agentum.afrodigital.dev/

Businesses need customer support that answers from their own documentation without building AI in-house. Built document ingestion (PDF, CSV, text, web scraping), custom-tone sections, and an embeddable chat widget, delivering a live platform any business can add to its website.

BioMatch | *AI-Powered Organ Donor Matching* – https://bio-match-tau.vercel.app/

Organ donor matching relies on manual compatibility checks that can take months. Founded and built an AI system that matches donors across multiple organs using population genetics data, tested end to end with Cypress and Selenium. Recognized as a top startup at Lead Business 2025, Ethiopia.

Technical Skills

Languages: TypeScript, Python, PHP, Ruby, Java, Scala, SQL, Bash

Backend & APIs: Node.js, NestJS, Bun, Express, Hono.js, Laravel, Ruby on Rails, FastAPI, REST, GraphQL, gRPC, tRPC, WebSockets, Microservices, OpenAPI/Swagger, Zod, OAuth, RBAC, Webhooks

Databases, Data & Messaging: PostgreSQL, MongoDB, Redis, SQLite, Elasticsearch, Supabase, BigQuery, Snowflake, Prisma, Drizzle, Kafka, BullMQ

Frontend: React, Next.js, TanStack, Redux, Tailwind CSS, shadcn/ui

Cloud, DevOps & Testing: AWS, Azure, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Linux, Nginx, Prometheus, Jest, Vitest, Supertest, Cypress, Selenium, Unit Testing

AI, Automation & Integrations: OpenAI API, Claude API, LangChain, RAG, MCP, AI Agents, LLM Training Data (SFT/RL), n8n, Zapier, Make, GoHighLevel, Telegram Bot API & Mini Apps, Google Workspace APIs

Engineering Practices: System Design, API Design, Performance Optimization, Code Review, Technical Mentoring, Agile/Scrum, Git

Education and Certifications

BITS College School of Systems and Technology — Addis Ababa, Ethiopia
Bachelor of Software Engineering — GPA: 4.0 / 4.0 — *2024 – Present*

Full Stack Web Development Certification — 2023 – 2024
Evangadi Tech

Programming Fundamentals — December 2024
Udacity

Certificate of Recognition — Top Startup (BioMatch) — November 2025
Lead Business 2025, SNV Ethiopia — pitched BioMatch to investors over a two-day program

Design Your Venture (DYV) — September 2026
Entrepreneurship Development Institute

Entrepreneurship Training Workshop (ETW) — September 2026
Empretec and Entrepreneurship Development Institute (EDI)

Languages

English (Fluent), Amharic (Native)